Magic Box

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A **Magic Square** in math is a square array of numbers in which each **row**, **column**, and **diagonal** of numbers have the **same sum**.

In other words, adding each row of numbers, each column of numbers, and each diagonal of numbers will give the same sum, called the magic constant.

```
import java.util.Scanner;

class MagicSquare {

   static int[][] createMagicSquare(int n) {

      if (n % 2 == 0){

         System.out.println("Magic square creation is supported only for odd n.");

         return null;

      }

      int[][] magicSquare = new int[n][n];

      int row = 0;

      int col = n / 2;

      magicSquare[row][col] = 1;

      for (int num = 2; num <= n * n; num++) {

         int newRow = row - 1;

         int newCol = col + 1;

         if (newRow < 0) newRow = n - 1;

         if (newCol == n) newCol = 0;
```

```
        if (magicSquare[newRow][newCol] == 0) {
            row = newRow;
            col = newCol;
        } else {
            row = row + 1;
            if (row == n) row = 0;
        }
        magicSquare[row][col] = num;
    }
    return magicSquare;
}
static boolean validateMagicSquare(int[][] matrix, int n) {
    int targetSum = 0;
    for (int j = 0; j < n; j++)
        targetSum += matrix[0][j];
    for (int i = 0; i < n; i++) {
        int rowSum = 0;
        for (int j = 0; j < n; j++)
            rowSum += matrix[i][j];
        if (rowSum != targetSum)
            return false;
    }
    for (int i = 0; i < n; i++) {
        int colSum = 0;
        for (int j = 0; j < n; j++)
            colSum += matrix[j][i];
        if (colSum != targetSum)
            return false;
    }
```

```java
    int diag1 = 0, diag2 = 0;
    for (int i = 0; i < n; i++) {
        diag1 += matrix[i][i];
        diag2 += matrix[i][n - 1 - i];
    }
    return diag1 == targetSum && diag2 == targetSum;
}
static void printMatrix(int[][] matrix, int n) {
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++)
            System.out.print(matrix[i][j] + "\t");
        System.out.println();
    }
}
public static void main(String[] args) {

    Scanner sc = new Scanner(System.in);
    while (true) {
        System.out.println("\n1. Validate Magic Square");
        System.out.println("2. Create Magic Square");
        System.out.println("3. Exit");
        System.out.print("Enter choice: ");
        int choice = sc.nextInt();
        switch (choice) {
            case 1:
                System.out.print("Enter size n: ");
                int n1 = sc.nextInt();
                int[][] matrix = new int[n1][n1];
                for (int i = 0; i < n1; i++){
```

```java
            for (int j = 0; j < n1; j++){
              System.out.print("Enter element [" + i + "][" + j + "]: ");
              matrix[i][j] = sc.nextInt();
            }
          }
          if (validateMagicSquare(matrix, n1))
            System.out.println("It IS a Magic Square.");
          else
            System.out.println("It is NOT a Magic Square.");
          break;
        case 2:
          System.out.print("Enter odd size n: ");
          int n2 = sc.nextInt();
          int[][] magicSquare = createMagicSquare(n2);
          if (magicSquare != null) {
            printMatrix(magicSquare, n2);
            if (validateMagicSquare(magicSquare, n2))
              System.out.println("Validation Passed.");
          }
          break;
        case 3:
          System.out.println("Exiting...");
          return;
        default:
          System.out.println("Invalid choice.");
      }
    }
  }
}
```

```
PS C:\Users\danis\Desktop\Striver-DSA> cd "c:\Users\danis\Desktop\Striver-DSA\" ; if ($?
) { javac MagicSquare.java } ; if ($?) { java MagicSquare }

1. Validate Magic Square
2. Create Magic Square
3. Exit
Enter choice:
```

```
PS C:\Users\danis\Desktop\Striver-DSA> cd "c:\Users\danis\Desktop\Striver-DSA\" ; if ($?
) { javac MagicSquare.java } ; if ($?) { java MagicSquare }

1. Validate Magic Square
2. Create Magic Square
3. Exit
Enter choice: 2
Enter odd size n: 3
8       1       6
3       5       7
4       9       2
Validation Passed.

1. Validate Magic Square
2. Create Magic Square
3. Exit
Enter choice:
```

```
va } ; if ($?) { java MagicSquare }

1. Validate Magic Square
2. Create Magic Square
3. Exit
Enter choice: 1
Enter size n: 3
Enter element [0][0]: 3
Enter element [0][1]: 4
Enter element [0][2]: 5
Enter element [1][0]: 3
Enter element [1][1]: 4
Enter element [1][2]: 3
Enter element [2][0]: 4
Enter element [2][1]: 2
Enter element [2][2]: 5
It is NOT a Magic Square.

1. Validate Magic Square
2. Create Magic Square
3. Exit
Enter choice:
```